Smart Multi-Sensor Wearable for Real-time Elderly Health Anomaly and Fall Detection

R26-IT-108

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specialization in Information Technology (IT)

Department of Information Technology

Sri Lanka Institute of Information Technology Sri Lanka

2026 March

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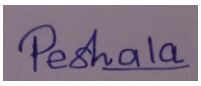
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DECLARATION

I declare that this proposal is my own work and does not incorporate, without acknowledgement, any material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of my knowledge and belief, it does not contain any material previously published or written by another person except where due acknowledgement is made in the text.

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The above candidate is carrying out this research for the undergraduate dissertation under my supervision.

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Date: 12.03.2026

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ABSTRACT

This study suggests a Neurological Stress and Pain monitoring device to a smart elderly healthcare system that is aimed at helping to detect hidden distress in older adults who might fail to express any pain or suffering verbally. The component is dedicated to the galvanic skin response (GSR) monitoring and electrodermal activity (EDA) analysis as physiological stress, agitation, and potential pain-related arousal indicators. The system is modelled as a non-invasive wearable mechanism and is developed based on an ESP32 microcontroller, skin conductance electrodes, and a built-in alert system.

The methodology consists of the continuous acquisition of physiological signals, preprocessing of the raw EDA signals (noise removal and tonic-phasic separation), the extraction of the feature of interest (mean, variance, amplitude, and peak count) and the classification of distress states with a Random Forest model. The system is also intended to be an Emotional Intelligence layer to the extended team architecture which can be integrated with heart rate and motion modules to enhance contextual accuracy and minimize false alerts.

One of the key advantages of the proposed system is that it utilizes edge-based processing, which provides privacy-aware local inference, low-latency response, and the ability to work in realistic elderly care environments. A working prototype that will be able to produce real-time caregiver alerts and deliver safer, more responsive, and intelligent elder care monitoring is expected.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
GSR	Galvanic Skin Response
EDA	Electrodermal Activity
SCR	Skin Conductance Response
SCL	Skin Conductance Level
ESP32	Espressif 32-bit Microcontroller
AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
SDG	Sustainable Development Goal
IoT	Internet of Things

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1. Introduction

1.1 Background and Context

The process of ageing in population has become a significant issue in healthcare on a global scale, particularly in matters that involve the ongoing monitoring of frail and aged adults suffering chronic conditions, cognitive impairment and less functional autonomy. One of the most at risk populations (among the aged) consists of those people who cannot express pain, stress, or discomfort in a verbal form. This can happen to those who are impaired by dementia, Alzheimer, stroke, or any neurological conditions or decline that decrease their ability to express themselves verbally and report their condition. In such people, symptoms that are clinically significant, like pain, fear, agitation, distress, or physiological stress can still go unnoticed until they develop into serious health outcomes. Consequently, this lack of verbal communication leads to a severe impediment in the management of timely intercession and safe care [2], [11].

Neurological stress and internal pain monitoring of elderly people is thus a very important health requirement. Stress and pain are not merely subjective, but also physiologically meaningful states, which can deteriorate the health outcomes when they are not controlled. Persistent distress in elderly patients may result in agitation, sleep disturbance, loss of mobility, worse recovery and worsening quality of life. Unobservable and unreported pain can also cause delay in treatment and unwarranted pain. This is more of an issue in care environments where few caregivers are required to be present to monitor several patients and are not always able to monitor all behavioral changes. In these populations, objective and continuous monitoring techniques with the capacity to detect distress-related physiological alterations in the absence of verbal reporting or untrustworthy verbal reporting are largely required [2], [9], [11].

The prospective solution to this issue is objective physiological monitoring. Rather than using self-report or visible symptoms or occasional clinical evaluation, wearable biosensors can keep track of body cues related to autonomic nervous system activities and emotional stimulation. It is one of the most pertinent indicators in this regard, the so-called galvanic skin response, or electrodermal activity. Electrodermal activity is a measure of variations in skin conductance due to the activity of the sympathetic nervous system and the presence of the sweat glands and has been extensively used to show changes in stress, emotional arousal, pain responses and has

been used to observe responses to pain [5], [6], [9], [10]. Since the electrodermal responses may be measured in a non-invasive and continuous process, which is achievable with wearable devices, they offer an objective channel by which the state of distress can be determined in older users who may not be able to provide direct information about his or her state.

This has resulted in the growing interest of affective computing in healthcare. An affective computing system is defined as the creation of systems capable of detecting, perceiving, and responding to human emotions or bodily conditions. Affective computing is particularly useful in elderly care, where emotional distress, agitation, and pain usually display insidious behavioral symptoms but are closely related by physiological means. Research on wearable-based affect recognition has demonstrated that physiological cues, including electrodermal activity, can be used to detect stress and affective conditions under non-controlled laboratory settings [3]. In the context of healthcare, these capabilities can be used to transform the internal discomfort that would be otherwise invisible into measurable signals to be responded to more promptly by caregivers and systems. In this regard, affective computing may be used as a type of physiological communication support to non-verbal or cognitively disabled elderly people [3], [11].

Despite these developments, existing wearable systems also have significant drawbacks. Most consumer wearables are primarily general wellness tracking, heart rate recording, sleep rating, or physical activity monitoring as opposed to clinically meaningful distress and pain monitoring [1], [2]. Stress detection is not always configured to suit older adults who may have a neurologically vulnerable elderly user population even where there is stress detection available. General-purpose devices also tend to give simplified output with lack of contextual interpretation, and it is hard to tell whether it is emotional distress and pain or exercise-related arousal or normal variation in physiology. Moreover, a significant number of systems depend on cloud connectivity or even smartphone-sized analysis, as opposed to embedded, privacy-sensitive local processing [1], [12]. It can lower responsiveness and bring up the issue of dealing with sensitive physiological data.

The significance of this issue is local and international. The rising ageing populations worldwide are driving up the need to find low cost, scalable, and ongoing care options that can mitigate the load of the hospitals, and enhance the quality of life of the elderly [1], [2]. Monitors of the elderly are ceasing to be restricted to hospitals and caring homes, it is more becoming home based, assisted living, and community-based care. Wearable devices used in these

settings can facilitate 24-hour surveillance without incurring the cost of having an actual clinical supervisor. On a local level, the applicability is also high in the care systems where the families and the caregivers and the nursing professionals are usually tasked with elderly individuals who have limited access to constant professional observation. An inexpensive wearable that can identify concealed stress and pain can thus make a significant contribution to domestic and institutional care of the elderly.

This study intends to overcome these issues by introducing Neurological Stress and Pain Monitoring components in a wider smart wearable healthcare platform used in the elderly care context. The component is supposed to capture electrodermal activity real-time with a wearable sensor and process the signal to detect physiological patterns of stress and pain. To be more specific, the system is aimed at isolating tonic and phasic elements of electrodermal activity, isolating meaningful features, and grouping distress-related states, using an embedded machine learning model. The processed outcome is then utilized to give timely notifications to caregivers in case of abnormal or prolonged distress. This enables the component to be a specialized emotional and neurological monitoring layer of the entire multi-sensor system [3]-[5], [9].

Embedded intelligence has been used to support the proposed approach. Recent work on edge AI and TinyML has demonstrated that machine learning can run on resource-constrained microcontrollers to perform real-time inference with less reliance on cloud deployment [12]. Within the framework of elderly care, it is very advantageous since the analysis of physiological data can be conducted on a local level, which, in turn, can decrease latency, enhance privacy, and ensure trustworthy functionality even in the case of poor network conditions. The integration of wearable electrodermal sensing with edge-based classification hence provides a convenient path towards real-time distress detection among non-verbal elderly users [6], [12].

The study also correlates with Sustainable Development Goal 3: Good Health and Well-being, as it focuses on guaranteeing healthy lives and well-being status of all, irrespective of their age. To the more responsive and safer care, a wearable system that aids the early detection of pain, agitation, and stress in the elderly is a direct contribution to safer and more responsive care. It assists in preventive care, better consideration of vulnerable groups and innovation in available health technologies. Using research, the study is also relevant to existing literature on wearable biosensing, affective computing, geriatric healthcare technology, and intelligent embedded

systems [1], [2], [12]. Thus, the suggested element is socially topical and academically based as a solution to a significant gap in the health monitoring of the elderly.

1.2 Literature Review

The body research on the topic of neurological stress and pain measurement with wearable devices covers several closely related topics such as wearable healthcare technologies, affect recognition, electrodermal sensing, objective pain assessment, elderly care technologies, and embedded intelligence. The combined results of these studies form a very solid basis to develop a wearable device capable of detecting non-verbal distress in ageing people by analyzing physiological data instead of verbal expressions to identify the distress.

The use of wearable health technologies in healthcare has gained a new significance in contemporary healthcare due to the ability to monitor over a long period outside the clinic. Dias and Cunha defined wearable devices as the main platforms where vital signs can be monitored continuously and non-invasively, which is why they were relevant to preventive and patient-centred care [1]. Olmedo-Aguirre et al. narrowed down this discussion to the context of the elderly populations, indicating that wearables can be used to support remote healthcare by enhancing safety, early intervention and health monitoring in the daily living conditions [2]. Such studies indicate that wearable is now a feasible and significant means of constant monitoring of the elderly. Nevertheless, they also imply that most of the existing systems are still general in nature and have not yet considered hidden psychological or pain-related conditions of geriatric populations completely [1], [2].

Affective computing has become one of the key research directions in interpreting physiological signals related to emotion and stress. A review of wearable-based affect recognition conducted by Schmidt et al. demonstrated that physiological signals, particularly electrodermal activity, are some of the most practical modalities to use in acknowledging stress-related and emotional conditions [3]. This justifies the application of electrodermal sensing in medical practice where there is no direct observation of emotional or neurological distress. This area was further enhanced by the WESAD dataset presented by Schmidt et al. that offers a popular multimodal benchmark of wearable stress and affect detection [4]. WESAD has gained significance due to its labelled physiological data, which may be utilized

to train and assess models to classify stress, which is particularly helpful in wearable machine learning studies.

The electrodermal activity itself is also a well-researched direct measure of the sympathetic nervous system. Sangrias et al. proved that it is possible to categorize calm and distress states with the use of electrodermal signals, proving that the alteration of the skin conductance is a significant indicator of the stress-related conditions [5]. Klimek et al. subsequently surveyed wearables to sense electrodermal activity in care settings and found that EDA-based devices can be utilized to assess perceived stress, although standardization and interpretability are still a challenge [6]. These studies have a direct relation to this proposal as they substantiate the view that GSR/EDA is not merely technically quantifiable in an easily wearable form, but also clinically significant in the care environment.

A significant analytical characteristic of electrodermal data is that it can be divided into tonic and phasic components. The tonic component, commonly known as skin conductance level, represents the changes in arousal that occur more slowly in the background, and the phasic component, also known as skin conductance response, represents the rapid transient responses to stimuli or acute stress. This difference is significant since the interpretation of meaningful stress data is frequently determined by not only absolute conductance values but also by the trend of temporary peaks and baseline drift. Strzinar and Lotric demonstrated that the performance of stress detection can be enhanced by considering the characteristics of electrodermal signals when measured on the wrist, as well as the properties of the signals in the frequency domain [7]. The value of structured processing EDA instead of using raw conductance values is supported by their findings.

Other contemporary research has delved deeper into sophisticated methods of data processing. Almadhor et al. examined the use of federated learning on wrist-based electrodermal monitoring to detect stress and showed that machine learning could improve the recognition of stress through wearable EDA signals [8]. Even though the computational method used is different to embedded TinyML deployment, this research remains applicable as it demonstrates that wearable electrodermal data could be used to support intelligent classification instead of threshold-based alerting only. It is also indicative of the general movement toward privacy-conscious physiological computing.

Another solid basis of the proposal is the literature on pain assessment. Aqajari et al. tested a pain measurement device on electrodermal activity in patients after surgery and demonstrated that EDA may be used in pain-specific surveillance of the objective [9]. In the same spirit, Posada-Quintero et al. indicated that the electrodermal responses can provide information about various degrees of pain stimulation in controlled conditions which reinforces the idea that skin conductance can be used as an informative measure of pain-provoking physiological response [10]. These results are quite applicable since it has been established that electrodermal sensing can be utilized in not only generic stress detection but also in checking internal discomfort and pain conditions that might otherwise be kept secretive.

This is especially significant in old and cognitively impaired groups. Melander et al. demonstrated that electrodermal activity has the potential to enhance detecting agitation on patients with dementia [11]. The research is particularly important to the current study as it directly handles a population where behaviour and communication do not always give a good indication of internal distress. It recommends that physiological surveillance could serve as an extra observational layer among susceptible aged individuals whose distress may be overlooked with conventional caregiving alone.

Lastly, proposed implementation model is supported by the literature on TinyML and embedded intelligence. Ray discussed TinyML as a promising direction of executing machine learning on low-power embedded systems and highlighted its usefulness in real-time, decentralized inference in real-life applications [12]. This applies greatly to healthcare as physiological monitoring can frequently be improved by near-time interpretation, small latency, and local processing without violating privacy. Embedded analysis can be used to provide real-time classification and create fast alerts, a feature especially desirable in wearable elderly care, instead of streaming the raw data to external systems.

Despite the overall positive improvement in these studies, there are still critical flaws. A great number of wearable systems do not target elderly neurological stress and pain monitoring in particular [1], [2]. Much of the research on affect recognition is conducted on the level of stress in healthy adults as opposed to distress in geriatric care [3], [4]. There is EDA monitoring research that confirms usefulness of signals, but it also identifies issues in standardization and contextual interpretation [6], [7]. The physiological validity of pain-related studies is upheld, but these studies are not always transferred to wearable real-time eldercare solutions [9], [10]. Likewise, technical opportunities are available through embedded AI research, but full

integration, edge based, geriatric distress-monitoring prototypes are still under-represented [12]. The restrictions are the reason why the current study is needed.

1.3 Research Gap

Although the field of wearable health monitoring is rapidly evolving, there is still a big gap in technology that will be able to detect the presence of neurological stress, hidden pain, and distress in users who cannot communicate reliably with their state, despite being old. Current wearable technology has been able to excel in fields like heart rate monitoring, activity tracking, fitness tracking as well as the overall stress estimation [1]-[3]. Yet, such systems are seldom adjusted to the specific problems of geriatric care, particularly of non-verbal or cognitively impaired older adults. Consequently, the existing technology environment fails to provide sufficient support to identify internal distress in the elderly populations where the behavioral indicators are too subtle or absent.

The first significant weakness is that most consumers and many clinical-proximate wearables do not provide a distress-monitoring interface. They also tend to record general wellness metrics like steps, heart rate, sleep, or general stress scores, but they do not explicitly test pain-related autonomic reactions, neurological distress, and agitation in cognitively vulnerable people [1], [2]. These systems are usually optimized to support healthy adult users, as opposed to the elderly with neurological decline, dementia or impaired communication. It implies that the interpretation logic in most existing devices is not clinically/contextually suitable to the target population of this proposal.

The second gap is an excessive dependence on few physiological indicators. Heart rate-only devices or general stress monitors cannot detect small signs of internal pain or emotional distress during autonomic arousal as it is multifaceted and cannot be understood in one indicator. The existing studies on wearable-based affect recognition indicate that electrodermal activity is among the most informative physiological cues that can be used to identify emotional arousal and stress [3], [5], [6]. Studies of pain-assessment also show that EDA has a significant positive relationship with discomfort and nociceptive stimulation [9], [10]. Thus, systems dependent mainly on heart rate or basic wellness scoring can overlook crucial sympathetic variation linked to pain, agitation or mental distress. This renders general-purpose trackers incompetent in special elderly distress monitoring.

The other significant gap is that there are no wearable systems aimed at objective distress monitoring in geriatric settings. Despite evidence on the utility of EDA in the analysis of stress and pain, most studies are at the signal interpretation phase, dataset-based classification, or controlled experiments [4]-[10]. Few systems have turned such findings into useful wearable devices that monitor distress in real time and facilitate the response of the caregiver. The article by Melander et al. is particularly relevant as it proves the practicality of EDA in agitation associated with dementia [11], but the rest of the field still requires integrated wearable technology to implement this principle in day-to-day eldercare.

An obvious lack of edge-based geriatric distress monitoring systems is also evident. Although TinyML and embedded inference have rendered local machine learning more a possibility [12], much of physiological monitoring architectures continue to rely on external computation, cloud services, or smartphone level analysis. This brings about problems of latency, dependence on connectivity, and privacy exposure. These restrictions are especially troublesome in elder care since the prompt interpretation and the lowest possible risk of data-sharing are both crucial. A distress-monitoring module capable of operating with physiological data directly on an embedded platform is hence a requirement that is not well developed in the current literature.

Another weakness is the absence of cross-referencing between electrodermal activity and other physiological or behavioral modules. The EDA is a strong tool, but its interpretation may be unclear. A high arousal can be a sign of pain, fear, emotional stress, physical activity or otherwise. Recent wearable research frequently examines electrodermal response alone [5]–[8], which may lower specificity in wearable applications. This is a severe limitation in the case of elderly healthcare. A stronger system ought to triangulate distress-relevant EDA patterns with constrained mobility and heart-rate data of other instrumentations that are part of the extended wearable platform. This enables the system to discriminate better e.g. between exercise related arousal, emotionally agitated, low mobility pain, or stress related to abnormal physiological state. Absence of such contextual integration is a research gap of meaning.

The originality of the suggested component is in answering these inadequacies directly. The proposed research suggests a wearable Neurological Stress and Pain Monitoring module based on GSR/EDA sensing, organized tonic-phasic analysis, feature extraction, and embedded machine learning to recognize distress in elderly users. The component, as opposed to general wearables, is tailored to non-verbal or communication-impaired elderly people. This is unlike the heart rate-only systems, which rely on a physiologically rich signal powerfully related to

autonomic arousal and pain response. In comparison with monitoring tools that are cloud-dependent, it is designed to apply edge-based local inference as a way of achieving privacy and fast response [11], [12]. Moreover, it is also designed to operate in a more generalized multi-sensor architecture in which contextual interpretation could be enhanced by connecting it to motion and cardiovascular components as opposed to isolated stress trackers.

This element is required due to insufficient service of the current solutions to serve as a practical, wearable, privacy-conscious, and elderly-specific mode of transforming invisible physiological distress into actionable care data. Older people who cannot articulate pain or stress are left susceptible to late assistance, unchecked suffering, and deteriorated health conditions. By bridging this gap, the proposed study offers an expert monitoring layer capable of enhancing caregiver awareness, promoting the execution of interventions sooner, and expanding the affective computing role into more geriatric care relevant settings [2], [3], [11].

2. Objective

2.1 Main Objective

This study aims to design, develop and test a wearable Neurological Stress and Pain Monitoring device in older people that continuously measures galvanic skin response (GSR) activity, interprets electrodermal activity (EDA) data, and sends real-time distress alerts using an embedded machine learning model running on an ESP32 based platform. The successful application of continuous physiological signal acquisition, tonic-phasic EDA analysis, feature extraction, Random Forest-based distress classification, and timely notifying a caregiver can help measure this objective in a wearable prototype setting [4], [5], [12].

2.2 Specific Objectives

To achieve the main objective, this study is guided by the following specific objectives:

1. **To develop a wearable physiological sensing module** capable of continuously measuring galvanic skin response from elderly users in a non-invasive manner, thereby

enabling real-time observation of autonomic nervous system activity associated with stress, discomfort, and emotional arousal [5], [6].

2. **To preprocess and analyse electrodermal activity signals** by separating tonic and phasic components to obtain meaningful physiological representations of sustained arousal and rapid sympathetic responses [6], [7]. This objective is important because raw GSR signals alone may not provide sufficient clarity for reliable distress interpretation.
3. **To extract relevant EDA features**, including skin conductance level (SCL), skin conductance response (SCR), peak-related characteristics, and statistical features, so that the wearable system can identify patterns associated with neurological stress and possible pain-related states [3], [7], [9].
4. **To design and implement a Random Forest-based classification model** that can distinguish normal and distress-related physiological conditions using processed EDA features and a research-supported training approach informed by wearable affect-detection studies such as WESAD [4], [8].
5. **To deploy the trained distress-detection model on an ESP32-based embedded platform** to support privacy-preserving, low-latency, and edge-based real-time inference suitable for wearable elderly healthcare environments [12].
6. **To generate timely caregiver-oriented alerts** whenever abnormal or sustained stress and pain-related physiological states are detected, thereby improving early intervention capability for elderly individuals who may be unable to communicate discomfort effectively [2], [11].

Collectively, these objectives support the creation of a practical and intelligent wearable solution that transforms hidden physiological distress into actionable care information for elderly monitoring contexts.

3. Methodology

3.1 Research Approach

The proposed study is an applied research approach since it is focused on addressing a practical healthcare challenge, namely, designing and developing a wearable technological solution. It is not purely theoretical research, but it is designed to result in a functional Neurological Stress and Pain Monitoring component, capable of aiding the elderly in the real world. In this case, applied research fits best since such a study aims to transform the available knowledge on electrodermal activity, stress monitoring, affect recognition, and embedded intelligence into a deployable system that meets the unmet demands of elderly, non-verbal or cognitively impaired people [1], [2], [11].

The research is also prototype-oriented. This method is appropriate since wearable healthcare research involves iterative design, hardware/software integration, signal collection, feature extraction, model implementation, and realistic testing. Previously, literature in wearable sensing and affective computing demonstrated that physiological monitoring systems should be vindicated in theory but also through practice and testing [3], [5]. As such, the study produces a prototype wearable device to record galvanic skin response, analyse electrodermal activity data, classify distress, and provide alerts to caregivers. The prototype-based approach will enable the research to determine technical feasibility, responsiveness, and applicability to elderly monitoring purposes.

3.2 General Methodological Framework

The research methodology is a chain of inter-connected steps back to sensing up to validation. The initial step is the identification and definition of the problem, during which the clinical and practical need to monitor neurological stress and pain in elderly care is converted to a series of technical goals. The second step is the system design where the sensing component, embedded processing workflow, classification logic, and alerting architecture are established. The third step involves hardware choice and integration, such as ESP32 controller, GSR sensor, electrodes of skin conductance, and alerting output devices.

The fourth stage is data acquisition and preprocessing. At this stage, constant electrodermal responses are recorded and processed to be analyzed by removing noise, decoupling tonic and phasic components, and isolating meaningful signal events like skin conductance responses. The fifth step is extraction and model development where descriptive physiological features are extracted and then utilized to train a classification model grounded in WESAD-inspired logic [4]. The sixth step is edge deployment and embedded inference where the classification logic is optimized to run on the ESP32. The last step is evaluation where the prototype is evaluated based on functional correctness, classification performance, alert responsiveness, and integration with the larger system. This hierarchical development process also guarantees the proposed component to be built systematically between sensing and real time validation [3], [12].

3.3 System Design Overview

The suggested system is set up as a wearable neurological stress and pain sensory input that acts as a layer of emotional intelligence on the larger elderly healthcare system. The system contains five main modules at a very high level, they are: the sensing module, preprocessing module, feature extraction and baseline module, classification module, and the alerting and communication module. These modules are sequential but interrelated to convert raw physiological activity into actionable warning information.

The sensing module records galvanic skin response in the user by recording, continuously, via galvanic skin response electrodes attached to a wearable GSR sensor. The acquired analogue signals are sent to the processing unit, where the preprocessing unit subtracts noise and decomposes the signal to meaningful parts. The feature extraction and baseline module then extract statistical and event-based features without losing a profile of the normal physiological behaviour of the user. The features are fed into the classification module where a Random Forest model is used to determine whether the observed pattern is either the norm state, stress-related arousal, or potential pain-related distress. Lastly, the alerting module creates local and care-giver-initiated warning when high-risk or persistent distress is identified.

This design is based on research that has revealed that electrodermal signals can be used to aid identification of stress, pain, and affective states when systematically processed instead of being interpreted in raw form [3], [5], [9]. The modular design also enables the proposed

component to be integrated with the wider team platform by exchanging classified distress data that can be later corroborated with heart rate and movement data to put the distress into context.

3.4 Sensor and Hardware Components choice

The hardware of the proposed system focuses on a small embedded system that can be integrated into a wearable system. The main processing device is the ESP32 microcontroller, which was chosen due to the adequate computing power it offers in performing lightweight signal analysis and model inference, as well as being able to operate with low power and to communicate wirelessly. These attributes render it the appropriate technology to edge-based physiological monitoring systems, particularly when the embedded inquiries need real-time inference [12].

The primary sensing device is a galvanic skin response sensor that is attached to skin conductance electrodes that are in contact with the skin of the user. This arrangement has been chosen as the electrodermal activity is among the most informative physiological indicators of sympathetic arousal associated with stress, agitation, and pain [5], [6]. The skin conductance electrodes are critical in obtaining stable bioelectrical changes related to sweat gland activity. Quality of contact is essential due to the changes in signal quality depending on sensor positioning and movement.

Hardware support hardware Support hardware is provided like a local alert device (vibration motor, buzzer or indicator light) to give immediate feedback when distress is detected. Such components are needed as it is required that the system should not only analyse physiological states but also communicate risk in a timely and practical manner. The chosen hardware thus enables real-time user or caregiver awareness, local analysis and continuous sensing within a wearable form factor.

3.5 Data Acquisition Process

Data acquisition process is structured in such a way that it can allow continuous sampling of galvanic skin response when the wearable is in use. When the user puts on the device, the skin

conductance electrodes continue touching the skin and the GSR sensor will start to gather physiological data at uniform sampling rates. Constant sampling is required, since physiological changes caused by distress can manifest slowly or intermittently and isolated spot samples are unlikely to reflect any significant dynamics in emotions or pain [3], [6].

The recorded signal is a record of changes in skin conductance through time, which is indicative of the work of the autonomic nervous system. Digitization of these readings is done by the built-in platform and stored temporarily to be processed further. Physiological signal capture process is to be non-invasive and the least intrusive to ensure the elderly users wear the device comfortably over long durations. The system, by generating GSR data in a time-series, produces a representation of physiological arousal over time, which can be analyzed in the future to capture long-term patterns, as well as short-term stress-inducing events [5], [11].

3.6 Signal Preprocessing Methodologies

Raw electrodermal signals when collected by wearable devices are susceptible to noise, motion artefacts, unstable electrode contacts, and environmental interference. Thus, signal preprocessing is a required methodological procedure, preceding feature extraction and classification. In this research, noise reduction methods are the starting point of preprocessing because it helps smooth the original GSR waveform and eliminate any arbitrary variations that are not linked to the physiological arousal. This aids in maintaining significant conductance distributions and enhancing signal stability [6], [7].

The electrodermal signal is then divided into tonic and phasic signals after noise removal. The tonic component is the more slowly changing base level of arousal, which can be defined simply as the skin conductance level, and the phasic component is the faster reactions to momentary stimuli, stress peaks, or sudden discomfort. This distinction is significant as the two components store valuable but distinct classes of information that can be utilized to monitor distress [7]. After isolating the phasic component, individual peaks of the skin conductance responses are located. These peaks are useful in that the frequency, amplitude, and time of occurrence may be used to signal acute arousal or stressful episodes.

During preprocessing, artifact handling is also done. Readings where there are bad contact, non-physiological spikes, or unrealistic changes of value are either corrected, smoothed or dumped. This helps in minimizing the chances of false classifications due to sensor instability

as opposed to actual physiological occurrences. The general preprocessing phase can be used to guarantee that the data that is fed to the classification model is physiologically significant and capable of wearable real-time analysis [5], [6].

3.7 Feature Extraction and Baseline Modelling

After preprocessing, the system derives several descriptive features of the cleaned electrodermal data. These characteristics are selected since they offer useful depictions of general arousal level and quick distress-related reactions. Statistical measures, including mean and variance, event-based measures, including SCR peak count, and amplitude-related characteristics, that characterize the strength of phasic responses, are extracted. These variables describe the behavioral pattern of the EDA signal together in a way that is applicable to machine learning-based classification [3], [7], [9].

Baseline modelling is factored in to enhance contextual interpretation of stress and pain conditions. Instead of constantly having to work with generalized physiological assumptions, the system has a user-specific baseline profile that represents the normal range of electrodermal behaviour of the individual. This is significant as there are physiological patterns of arousal which could vary significantly across users, especially in ageing populations with varying neurological and emotional backgrounds [6], [11]. The system can compare current features with a known personal baseline to help determine what is true distress as opposed to normal physiological fluctuation. This method enhances interpretability and decreases the chances of over-sensitive alertness.

3.8 mechanism of Distress Detection

The model that forms the basis of the distress detection mechanism is a random forest classification model that is developed to classify electrodermal patterns as pertaining to relevant physiological states. Random Forest is preferred due to its strength, interpretability, and the ability to work with structured groups of features obtained by processing physiological measures. The model uses the extracted EDA features and gets a classification result of whether

the pattern identified represents a normal state, stress-related arousal, or potential pain related distress.

The rationale behind the development of models is the organization of the WESAD dataset and the associated wearable stress-detection studies [4]. WESAD offers labelled physiological data in the analysis of stress and affective states and can be, thus, used as a good source to design a feature-driven classification. Though the final prototype will be based on this particular elderly-care application, the classification approach is based on the proven fact that wearable physiological characteristics can be used to facilitate machine learning-based affect recognition [3], [4]. The point of the mechanism is not to diagnose a particular medical condition, but to recognize notable physiological distress conditions which warrant attention and intervention.

3.9 Edge Processing and Embedded Intelligence

One important methodology aspect of the proposed system is that it uses edge processing and embedded intelligence. Once the classification model is built and tested, it is then optimized to run on the ESP32 platform. This can include making models simpler, less consumptive of memory and making inference low-computationally overhead. The aim is to enable real-time physiological interpretation in the wearable device itself and not through cloud-based analysis [12].

This design offers several benefits. First, local inference saves latency, which means that the system can react faster to the detection of distress. Second, it enhances privacy since raw physiological data does not have to be constantly sent to external systems. Third, it facilitates increased operational autonomy in settings where there is a low network connection. Edge intelligence is thus not only an optimization strategy of the technical nature but a design option appropriate to the practical demands of elderly healthcare monitoring [12].

3.10 Alert Generation and Notification Process

When the classification mechanism detects a distress-related state, the system triggers its alert generation mechanism. The lower level of response is a local haptic notification like vibration, which gives immediate notification via the wearable device itself. This local feedback may

prove helpful in confirming the system itself as well as in attracting attention to a potential abnormal physiological state.

The second tier of response is the care giver-initiated notification. An alert message or distress status is sent as a summary to a connected caregiver application or monitoring interface, enabling caregivers or healthcare personnel to view the condition and make additional action as needed. Critical alert logic determines the notification process by distinguishing mild, moderate, and severe distress patterns. An example is that transient physiological effects can cause a low-priority status, whereas sustained or acute abnormal patterns can cause immediate escalation. The significance of this tiered logic is that it can be used to minimize unnecessary alarms load and maintain sensitivity to potentially serious events [2], [11].

3.11 Prototype Development

The process of prototype development is accomplished by incorporating hardware, firmware and embedded analytical logic. The initial process is the connection of the sensing hardware, which contains the GSR sensor, electrodes of skin conductance, ESP32 control board, and local alert elements. The firmware development follows afterwards, and here the embedded software is coded to control signal acquisition, preprocessing, feature extraction, classification, and communication. Subsequently, all the modules are assembled into a wearable prototype that can be used to constantly monitor physiological conditions.

The prototype is built in stages in such a way that every functional unit can be tested separately before complete integration. Such a staged method will be useful in detecting hardware or software failures in a timely manner and enables tuning of the data flow and alerting logic. Since wearable healthcare systems must be stable and usable and functional intelligence should be implemented, practical factors like sensor contact, portability, and physical comfort are also considered during the assembly of prototypes [1], [6].

3.12 System Evaluation Methods

The proposed component is tested using several complementary testing techniques. Functional testing is performed to ensure that the system can properly obtain GSR data, pre-process the

signal, extract features, perform classification and produce alerts. The classification testing is conducted to test the reliability of the Random Forest model to detect the distress-related physiological states, using the given set of features and the training logic.

Latency testing is employed to identify the speed with which the system can react from the moment the physiological data has been recorded to the release of an alert. This becomes particularly noteworthy with real-time monitoring applications in which a response that is slow in coming diminishes utility. It also features integration testing to prove that the wearable unit can talk to outside caregiver interfaces as well as work as an extension of the larger multi-sensor elderly healthcare platform. These testing techniques offer an objective evaluation of the performance of the algorithms as well as the realistic functioning of the systems [4], [12].

3.13 Performance Measurement Criteria

Accuracy, precision, F1-score, latency, and false alerts reduction evaluate the performance of the proposed system. The overall percentage of correct classifications is measured by accuracy and precision by how well the system can identify distress states in the absence of too many false positives. The reason why F1-score is incorporated is that it presents a balanced value of accuracy and sensitivity, and thus it is reasonable to use when the precision and sensitivity of the classification performance require consideration [4], [8].

Measuring latency is the duration of time it takes the system to process the incoming physiological data and issue an alert. This condition is critical since the usefulness of a wearable distress-monitoring system is contingent on the promptness of a reply. False alert reduction is also considered as a valued measure since regular misleading notifications may lower the confidence towards the system and accidently overload caregivers. The combination of these criteria offers a realistic and research-based foundation for assessing technical precision and practicality.

3.14 Ethical and Safety Considerations

The ethical and safety aspects are at the core of the proposed methodology since the system must deal with physiological signs that could provide sensitive data about the emotional or

neurological condition of the user. This is why privacy preserving local processing is used as a design principle wherever possible. The system saves sensitive physiological information from exposure to the outside environment by processing electrodermal data on the embedded platform instead of sending all the raw signals elsewhere [12].

The system can also be worn non-invasively. The reason why GSR sensing using skin conductance electrodes is chosen is because they can obtain desired physiological activity without invasive procedures or painful measurement techniques [5], [6]. Moreover, the system is placed as an aid not a diagnostics tool. It is not to make definitive clinical diagnoses but to recognize patterns that are in keeping with stress, pain, or distress and to give early warnings to the caregivers. This restriction is significant since physiological arousal may possess a variety of sources and needs to be considered in the context of more comprehensive care [9], [11]. Implementation on an ethical basis must thus be transparent, protect privacy and communicate the intended use of the device carefully.

4. High-Level System Architecture

The Neurological Stress and Pain Monitoring component is specialized and proposed as a wearable subsystem as part of the larger smart elderly healthcare platform. It is mainly meant to detect the presence of hidden physiological indicators of emotional distress, neurotic stress, and potential pain among elderly users who might fail to convey their state of being unambiguously. On a high-level scale, the architecture is structured into sensing, preprocessing, inference, alerting, and caregiver communication. In the general team solution, this element acts as the Emotional Intelligence layer, since it transforms electrodermal activity into useful distress-related information that can aid in previous intervention and more responsive care [3], [5], [11].

The architecture will start with the sensing layer where galvanic skin response is constantly measured using electrodes in contact with the wearable GSR sensor. It is an indicator of activity of the sympathetic nervous system, so it will be applicable to the observation of stress, agitation, and pain-induced arousal [5], [6]. The perceived data is passed directly to the embedded controller where the local processing and decision support is done.

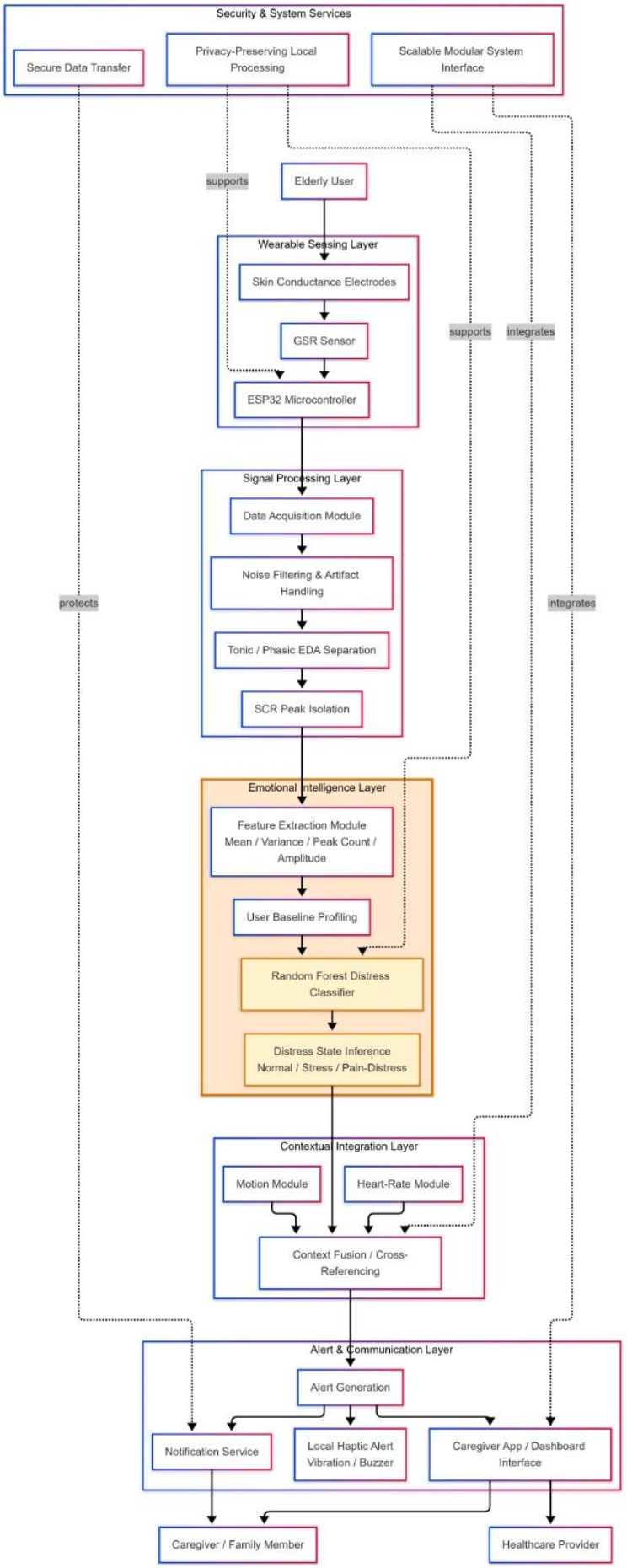


Diagram 1- High-level architecture diagram

The second layer is the preprocessing layer which preprocesses the raw signal to enable meaningful analysis. Since wearable electrodermal signals are prone to motion artefacts, unstable contacts and environmental noise, preprocessing is necessary to enhance reliability [6], [7]. The system in this layer does noise filtering, signal stabilization, artefact management, and tonic and phasic EDA separation. The tonic activity assists in interpreting the slower arousal patterns whereas the phasic activity assists in the interpretation of faster responses like peak of skin conductance response. This preprocessing phase guarantees that the information sent to inference mechanism is physiologically significant and can be used in detecting wearable distress.

The central analytical part of the architecture is the inference layer. This is where the element is most evidently perceived as the Emotional Intelligence layer. Relevant signal features are extracted (after preprocessing) and compared to learned or expected behavioral patterns (mean, variance, amplitude and SCR peak count). The features then undergo the incorporated Random Forest classification model that identifies the next option behaviour the user is either in a normal condition, or is aroused due to stress, or may be in distress/pain [4], [8]. The inference layer is the decision intelligence of the component by converting the physiological measurements into decodable risk data.

One of the strongest aspects of architecture is that it interacts with the larger team system, particularly the heart-rate module and the motion module. Though this aspect concentrates primarily on GSR / EDA, distress interpretation is enhanced when the electrodermal results are looked at in conjunction with cardiovascular and movement results. As an example, high levels of electrodermal activity at high motion can indicate physical activity or temporary exertion, and high levels of electrodermal activity with low motion and abnormal heart-rate behaviour are possible indicators of pain, fear, agitation or internal discomfort. Such contextual cross-referencing enables ambiguity reduction and enhances the detectability of real-world applications. As such, although the current component makes a localized inference of its own, it is also intended to share classified distress information with the broader platform to facilitate more multi-sensor interpretation [3], [11].

The inference process is followed by the alerting layer. The system will send a real-time warning when it recognizes a state of high distress. This can take the form of a local haptic notification by vibration or some other on-device cue to suggest that the presence of abnormal physiological stress has been identified. Meanwhile, the classification output can be forwarded

to the remote awareness caregiver communication layer. It is this sensing preprocessing inference alerting loop which describes the primary functional route of the system and is used to guarantee the architecture allows physiological interpretation to occur in time and not just passive data collection.

The last layer is the caregiver interface layer, or the interface that shows warning messages or a summary of distress information to the caregivers, family members, or other appropriate monitoring personnel. This can be done via a related mobile app or dashboard interface. The objective of this layer is never to give pure physiological complication but to relay actionable and intelligible caution returns. This contributes to the practicality of the system to ordinary care-giving environments where caregivers need rapid awareness as opposed to continuous manual interpretation of bio signals [2], [11].

Security is a significant architectural factor. The component processes sensitive physiological information and as such, privacy must be preserved. This is why architecture focuses on local bio signal analysis of the embedded device, as always opposed to the transmission of raw bio signal data. Edge-based inference minimises the unnecessary publicity of individual health-based signals and aids with a safer processing of user information [12]. External communication of only alert status or summarized information where necessary enhances confidentiality and makes less use of external infrastructure.

The architecture also supports scalability in terms of modular design. Future development can be expanded on each layer of the system. Other features of physiological description, better classifier, better caregiver interface, or stronger connection to other modules can be added without rearchitecting the whole structure. It implies that the suggested Emotional Intelligence layer is not only appropriate to the modern goal of neurological stress and pain monitoring but also offers a scalable framework to have in place in the future in the context of a larger smart elderly monitoring ecosystem [1], [12].

5. User Requirements

Table 5.1 Stakeholder Identification

Stakeholder	Role in the System	Main Needs / Expectations	Relevance to the Component
Elderly Users	Primary end users who wear the monitoring device	Comfortable wearable use, non-invasive sensing, reliable distress detection, minimum disturbance, quick warning support	The component continuously monitors neurological stress and pain-related physiological changes from the elderly user through GSR sensing
Caregivers / Family Members	Secondary users who receive warning information and respond to abnormal situations	Timely alerts, easy-to-understand notifications, reduced false alarms, remote monitoring capability	They depend on the component to identify hidden distress and notify them for quick intervention
Healthcare Providers / Nursing Staff	Professional users who may review alerts and monitor summaries for care support	Clinically useful alerts, trustworthy status information, support for care decisions, simple interpretation of results	They use the output of the component as a supportive monitoring aid in elderly care environments

Table 5.2 Functional Requirements

Req. No.	Functional Requirement	Description
FR-01	GSR Sampling	The system shall continuously acquire galvanic skin response data from the elderly user through skin conductance electrodes and the GSR sensor.

FR-02	Physiological Signal Capture	The system shall capture electrodermal activity data in real time and forward the sensed data to the embedded controller for processing.
FR-03	EDA Decomposition	The system shall preprocess the GSR signal by separating tonic and phasic electrodermal activity components for meaningful physiological analysis.
FR-04	Noise and Artifact Handling	The system shall reduce noise and handle motion-related artifacts or unstable signal readings before feature extraction.
FR-05	SCR Peak Detection	The system shall isolate skin conductance response peaks from the phasic component for distress-related event analysis.
FR-06	Feature Extraction	The system should extract relevant EDA features such as mean, variance, amplitude, and peak count from the processed signal.
FR-07	Baseline Profiling	The system should maintain a user-specific physiological baseline to improve personalized distress interpretation.
FR-08	Distress Classification	The system shall classify the user's physiological state into normal, stress-related, or pain/distress-related categories using the embedded Random Forest model.
FR-09	Contextual Alerting	The system shall support contextual distress interpretation by allowing cross-reference with heart rate and motion modules of the main team system.
FR-10	Local Alert Generation	The system shall generate an immediate local alert, such as vibration or buzzer output, when abnormal distress patterns are detected.
FR-11	Caregiver Notification	The system shall send warning notifications or summarized distress information to the caregiver interface or connected application.
FR-12	Integration with Main Team System	The system shall communicate classified distress outputs to the broader elderly healthcare platform for coordinated monitoring and response.

Table 5.3 Non-Functional Requirements

Req. No.	Non-Functional Requirement	Description
NFR-01	Reliability	The system should provide stable and dependable physiological monitoring during continuous wearable operation.
NFR-02	Low Latency	The system should process sensed data and generate alerts within a short response time suitable for real-time elderly monitoring.
NFR-03	Privacy	The system should preserve user privacy by performing as much processing as possible locally on the embedded device and minimizing unnecessary external transmission of raw physiological data.
NFR-04	Calibration Support	The system should allow initial calibration or baseline establishment for each elderly user to improve interpretation accuracy.
NFR-05	Accuracy	The system should achieve acceptable classification accuracy in identifying normal, stress, and distress-related physiological states.
NFR-06	Uptime	The system should remain operational for extended monitoring periods with minimum interruption.
NFR-07	Wearability	The device should be lightweight, non-invasive, and comfortable enough for practical use by elderly users.
NFR-08	Scalability	The component should be capable of integrating with additional sensors, modules, or future system upgrades without major redesign.
NFR-09	Maintainability	The firmware and hardware setup should be structured in a way that supports testing, updating, and future improvements.
NFR-10	Usability	Alerts and monitoring output should be simple and understandable for caregivers and healthcare staff.

Table 5.4 Summary of User Requirements by Stakeholder

Stakeholder	Key Functional Requirements	Key Non-Functional Requirements
Elderly Users	GSR sampling, signal capture, local alert generation	Wearability, reliability, privacy, uptime
Caregivers / Family Members	Caregiver notification, contextual alerting, integration with main system	Usability, low latency, accuracy, reliability
Healthcare Providers / Nursing Staff	Distress classification, monitoring support, system integration	Accuracy, calibration support, maintainability, scalability

Table 5.5 Expected Test Cases

Test Case ID	Test Case	Short Description	Expected Result
TC-01	GSR Signal Acquisition Test	Check whether the GSR sensor captures skin conductance data correctly from the user.	Valid GSR readings are received continuously.
TC-02	Signal Preprocessing Test	Verify filtering, noise reduction, and artifact handling of raw EDA signals.	Cleaned signals are produced with reduced noise.
TC-03	Tonic/Phasic Separation Test	Test whether the system separates tonic and phasic EDA components properly.	Tonic and phasic outputs are generated correctly.
TC-04	SCR Peak Detection Test	Check whether SCR peaks are identified from the phasic signal.	Relevant SCR peaks are detected accurately.
TC-05	Feature Extraction Test	Verify extraction of mean, variance, amplitude, and peak count features.	Correct feature values are generated from processed data.
TC-06	Distress Classification Test	Test whether the Random Forest model classifies normal and distress states correctly.	System outputs correct classification label.

TC-07	Contextual Integration Test	Verify whether EDA results can be combined with heart rate and motion module outputs.	Context-aware decisions are generated successfully.
TC-08	Local Alert Test	Check whether vibration or buzzer is triggered during abnormal distress detection.	Immediate local alert is activated.
TC-09	Caregiver Notification Test	Test whether alert information is sent to the caregiver app/dashboard.	Caregiver receives warning notification successfully.
TC-10	Latency Test	Measure how quickly the system responds from sensing to alerting.	Alert is generated within acceptable response time.
TC-11	Baseline Calibration Test	Verify whether the system stores and uses user baseline values correctly.	Personalized baseline is established and applied.
TC-12	End-to-End System Test	Test the complete flow from sensing to caregiver alert.	Full system works correctly without failure.

6. Commercialization Plan

The Neurological Stress and Pain Monitoring component proposed has high commercialization potential in the emerging elderly healthcare technology market. The main target market segment includes old age care facilities, assisted living facilities, geriatric care units of hospitals, and home care settings where vulnerable elderly citizens must be always observed. This is particularly applicable to older adults who possess cognitive impairment, neurological diagnoses, restricted mobility, or have diminished pain, stress, or discomfort verbalization. The growing need of practical wearable health monitoring systems is rising both locally and globally, owing to the ever-growing ageing populations, and the fact that the number of caregivers has been limited.

This product will majorly target the following customer groups: elderly care institutions, home-care service providers, families that take care of the older adults at home, and healthcare

providers or nursing staffs that monitor elderly patients. Care homes can embrace the system to enhance safety and early identification of latent suffering of the residents, and families can embrace the system as an assistive technology to monitor the elderly relatives remotely. Medical institutions and nursing care might consider it as an alternative physiological observation device among patients with a low ability to express their internal discomfort clearly.

The main pitch of the proposed system is that the proposed system is a non-invasive, wearable, real-time distress monitoring system that detects the latent neurological stress and physiological alterations in the body that precursors to more severe health conditions before it develops further. This system, in contrast to general wellness trackers which primarily monitor activity or heart rate, interprets electrodermal-based distress specifically and is an Emotional Intelligence layer on top of a bigger smart elderly care platform. It also enhances privacy, response time and can be used in an environment with limited connectivity due to its edge-based processing.

One of the potential pricing options is to sell the device and offer it as a service. As an illustration, the wearable device can be sold as a single hardware albeit to institutions or personal customers, and the caregiver dashboard, monitoring support, and software updates can be offered on a monthly or annual subscription. This enables the product to both raise upfront and repeat revenue. A relatively affordable package would be more appealing to the local market entry, particularly to small elderly care homes and middle-income home-care users.

The competitive advantage of the system is its specialization. There are numerous existing wearable products that are aimed at general activity tracking, fitness, or bare vital sign monitoring, but few offer any specific physiological intelligence related to distress that can be used by elderly users who might be non-verbal or cognitively impaired. The combination of the GSR-based stress tracking, contextual communication to the heart rate and motion modules, and privacy-preserving edge inference makes such a component unique in the market.

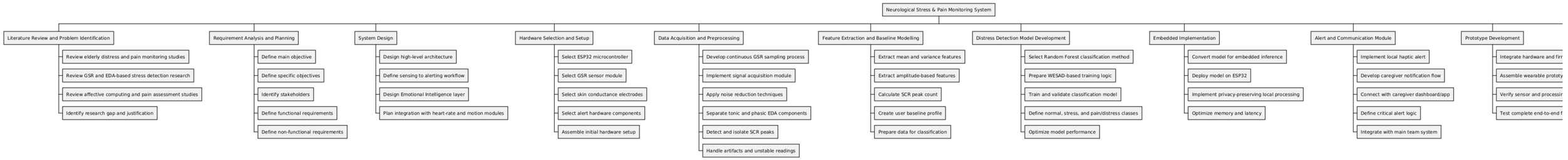
The most probable revenue model will encompass the sale of the hardware, subscriptions to the caregiver app, institutional licensing, and potential collaborations with healthcare providers or eldercare service organizations. The product might also be extended in the future to a higher-end-level multi-sensor elder monitoring product with predictive analytics, cloud reporting, remote patient dashboard, and hospital grade care integration.

7. Budget and Justification

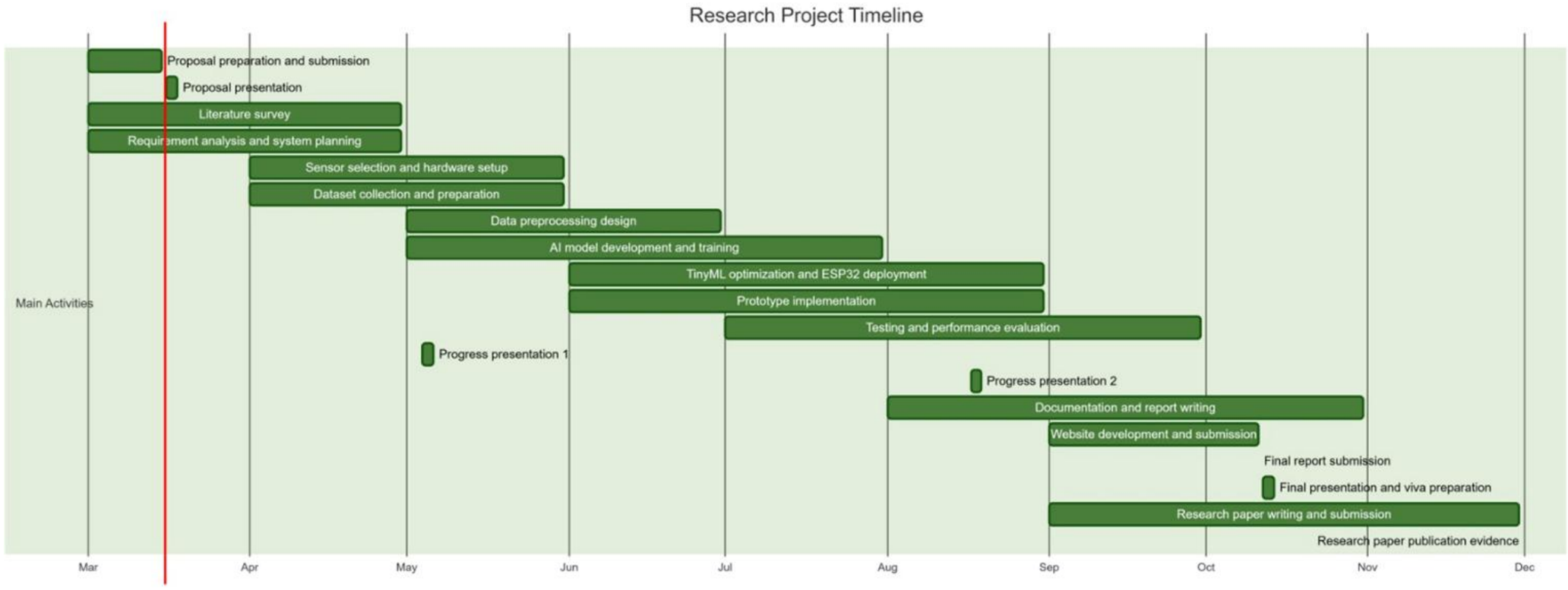
Table 7.1 Budget and Justification

Item Category	Item / Resource	Estimated Cost (LKR)	Justification
Hardware	ESP32 Microcontroller	3,500	Required as the main embedded controller for data acquisition, local processing, and communication.
Hardware	GSR Sensor Module	4,500	Needed to capture electrodermal activity for neurological stress and pain monitoring.
Hardware	Skin Conductance Electrodes	2,000	Essential for collecting physiological skin conductance signals from the elderly user.
Hardware	Alert Components (vibration motor / buzzer / LEDs)	1,500	Required for local real-time warning generation when distress is detected.
Hardware	Wiring, breadboard, connectors, wearable assembly materials	2,500	Needed for circuit integration, prototype assembly, and wearable implementation.
Software / Tools	Arduino IDE / Embedded programming tools	0	Open-source tools are sufficient for firmware development and ESP32 programming.
Software / Tools	Python / ML libraries	0	Open-source tools can be used for preprocessing, feature extraction, and Random Forest model development.
Dashboard / App Integration	Caregiver dashboard or mobile app prototype integration	8,000	Needed to develop or simulate a simple caregiver interface for alert display and monitoring.
Testing Resources	Test data preparation and validation resources	4,000	Required for prototype evaluation, sample data handling, and repeated functional and classification testing.
Human Effort	Design, development, integration, testing, and documentation effort	20,000	Represents the estimated effort value of building the prototype, evaluating performance, and preparing the research output.
Total Estimated Budget		**46,000**	

8. Work Breakdown Structure (WBS)



9. Gantt chart



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